Notes: Gödker, Jiao & Smeets (RFS, 2025)
Investor Memory

Contents

1 Big picture

- **Question.** How does memory affect investors' beliefs, reinvestment decisions, and overconfidence after they experience investment outcomes?
- **Main mechanism.** Investors exhibit a positive memory bias: they remember more gains and fewer losses than actually occurred.
- **Core design.** The experiments compare memory elicited immediately after observing outcomes with memory elicited one week later.
- **Main result.** After one week, subjects recall about 23% more positive outcomes and about 10% fewer negative outcomes than actually occurred.
- **Belief, choice, and confidence effects.** Biased memory generates optimistic beliefs, excessive reinvestment, and overconfidence.
- **Economic takeaway.** Memory provides a cognitive microfoundation for why investors learn more from past successes than from past failures.

2 Data and measurement

2.1 Experimental overview

The evidence comes from three experiments. Experiment 1 studies memory bias, belief formation, and reinvestment. Experiment 2 studies mechanisms using high recall incentives and no-choice treatments. Experiment 3 studies confidence and overconfidence.

Interpretation: The design follows the full chain from experience to memory to beliefs and subsequent financial behavior.

2.2 Investment environment

Subjects choose between a risky stock and a bond, or between risky stocks. In experiment 1, the risky stock is good or bad with equal prior probability. A good stock generates positive outcomes with probability 60%; a bad stock generates positive outcomes with probability 40%. Subjects observe 12 outcomes.

Interpretation: The data-generating process is known, so subjective beliefs can be compared to objective Bayesian benchmarks.

2.3 Treatment conditions

The main treatments are Delay, Immediate, NoRecall, Reminder, HighStakes, and NoChoice. Delay creates a one-week gap between outcome observation and memory elicitation. Immediate elicits recall right away.

Interpretation: The Delay–Immediate comparison isolates memory processes that occur after initial information acquisition.

2.4 Memory, belief, reinvestment, and confidence measures

$$MB_i^+ = RecalledPositive_i - ActualPositive_i, \qquad MB_i^- = RecalledNegative_i - ActualNegative_i.$$

Belief distortion equals subjective posterior belief minus the objective Bayesian posterior. Suboptimal reinvestment occurs when the subject chooses the stock even though the Bayesian benchmark favors the bond. Confidence is measured by points bet on the optimality of one's own investment choice.

Interpretation: These measures connect recall distortions to beliefs, decisions, and overconfidence at the individual level.

3 Models and empirical specifications

3.1 Bayesian benchmark

$$Pr(Good \mid n_+, t) = \frac{Pr(n_+ \mid Good)Pr(Good)}{Pr(n_+ \mid Good)Pr(Good) + Pr(n_+ \mid Bad)Pr(Bad)}.$$

Interpretation: This benchmark converts the observed outcome sequence into the objectively correct belief that the stock is good.

3.2 Belief regressions

$$SubjectiveBelief_i = \alpha + \beta_1 MB_i^+ + \beta_2 MB_i^- + \gamma ObjectivePosterior_i + \delta_s + \varepsilon_i.$$

Interpretation: The coefficients on memory bias show whether recall distortions shift beliefs even after controlling for the objective information.

3.3 Reinvestment and confidence regressions

$$Pr(SuboptimalReinvest_i = 1) = \Lambda(\alpha + \beta_1 MB_i^+ + \beta_2 MB_i^- + \delta_s),$$
$$PointsBetSuboptimal_i = \alpha + \beta MB_i^+ + \varepsilon_i.$$

Interpretation: These specifications test whether biased memory causes excessive repetition of past choices and overconfidence about one's own investment ability.

4 Identification and empirical design

The core identification strategy is a randomized timing comparison. Subjects in Delay and Immediate conditions observe comparable information, but only Delay subjects must rely on memory after one week. The gap between Delay and Immediate therefore identifies the effect of memory processes. The Bayesian benchmark identifies whether beliefs and choices are objectively distorted. HighStakes and NoChoice treatments test whether the bias is motivational rather than passive forgetting.

Interpretation: The strongest evidence comes from the combination of randomized timing, direct memory elicitation, objective Bayesian benchmarks, and mechanism treatments.

5 Tables and figures

5.1 Table 1: Overview of treatment conditions across three experiments; Figure 1: Subjective memory bias

Read: Table 1 maps treatments to timing. Figure 1 shows that delayed subjects remember more positive outcomes and fewer negative outcomes than actually occurred.

Detailed interpretation: The table clarifies the source of identification: timing and reminder variation. The figure shows that recall is directionally biased, not merely noisy.

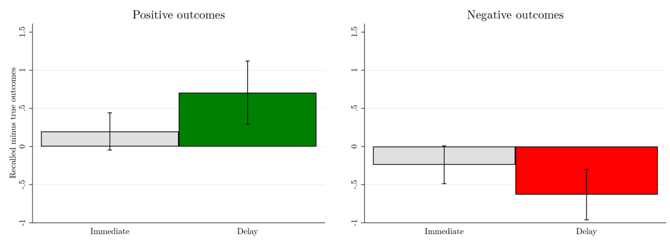
Economic message: Memory is already biased before the subject forms beliefs or makes the reinvestment decision.

Table 1
Overview of treatment conditions across three experiments

Treatment	Observation of outcomes	Memory elicit.	Reminder	Belief and/or choice elicit.	Exp.
Delay	Week t	Week t+1	No	Week t+1	1, 2, 3
Immediate(1)	Week t	Week t	No	Week t	1, 2
Immediate2	Week t+1	Week t+1	No	Week t+1	1
NoRecall	Week t	No	No	Week t+1	1
Reminder	Week t	No	Week t+1	Week t+1	3

This table provides an overview of the treatment and control conditions of the experiments.

(a) Table 1: Overview of treatment conditions across three experiments.



(b) Figure 1: Subjective memory bias.

5.2 Table 2: Subjective memory bias across treatments; Figure 2: Subjective beliefs

Read: Table 2 quantifies the gain-overremembering and loss-underremembering bias. Figure 2 compares subjective beliefs to Bayesian posteriors.

Detailed interpretation: The visual and regression evidence show that recall distortions are converted into optimistic beliefs. The subjective belief curve lies above the 45-degree Bayesian benchmark in the Delay condition.

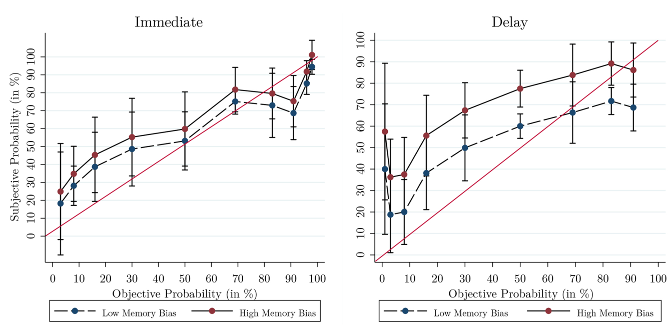
Economic message: The memory bias affects the information set investors use when updating beliefs.

Table 2
Subjective memory bias across treatments

	Immediate		*Delay*		Difference
Observed outcomes	Mean (N = 78)	t -test	Mean (N = 74)	t -test	t -test
Positive outcomes	0.27	$p = .058$	0.89	$p = .000$	$p = .018$
Negative outcomes	−0.28	$p = .037$	−0.73	$p = .000$	$p = .038$

This table displays subjects' memory bias in the *Delay* and *Immediate* condition in experiment 1. Memory bias is estimated at the individual level by subtracting the actual individually observed number of positive (negative) outcomes from subject's recalled number of the stock's positive (negative) outcomes. The table reports mean values and t-test results against the null hypothesis that the memory bias is zero and t-test results of the difference in means between our conditions *Delay* and *Immediate* for subjects who invested in the stock. Note that the sample is restricted to subjects who invested in the stock.

(a) Table 2: Subjective memory bias across treatments.



(b) Figure 2: Subjective beliefs.

5.3 Table 3: Memory-based beliefs

Read: Overremembering positive outcomes raises the subjective probability that the stock is good; underremembering losses also increases optimism.

Detailed interpretation: Table 3 is the main individual-level link between memory and beliefs. It conditions on the objective posterior, so the memory coefficients compare subjects with the same actual evidence but different remembered evidence.

Economic message: Biased recall is a mechanism for optimistic investor beliefs.

Table 3
Memory-based beliefs

	(1) Subj. prob.	(2) Subj. prob.	(3) Belief dist.	(4) Belief dist.
Memory bias (pos. out.)	6.196***		0.424***	
	(0.92)		(0.05)	
Memory bias (neg. out.)		–7.958***		–0.525***
		(1.02)		(0.06)
Objective prob.	0.684***	0.701***		
	(0.05)	(0.05)		
Constant	16.751***	15.204***	–0.185	–0.223
	(5.16)	(5.00)	(0.27)	(0.26)
Session	Yes	Yes	Yes	Yes
N	188	188	182	182
R^2	.52	.55	.31	.35

This table contains the coefficients and standard errors (in parentheses) of OLS regressions in which the dependent variable is the subjective posterior belief that the stock is the good stock (from 1 to 100), *Subj. prob.*, or subjects' belief distortion measured by the difference between the posterior log-likelihood ratios of subjects' elicited probabilities and the objective Bayesian probabilities, *Belief dist.* from experiment 1. *Memory bias (pos. out.)* represents subject's memory bias for observed positive outcomes. This variable is estimated at the individual level by subtracting the actual individually observed number of positive stock outcomes from subject's recalled number of the stock's positive outcomes. *Memory bias (neg. out.)* represents subject's memory bias for observed negative outcomes. This variable is estimated at the individual level by subtracting the actual individually observed number of negative stock outcomes from subject's recalled number of the stock's negative outcomes. *Objective prob.* is the value of the objective Bayesian probability that the stock is the good stock (from 1 to 100). We control for session fixed effects. * $p < .1$; ** $p < .05$; *** $p < .01$.

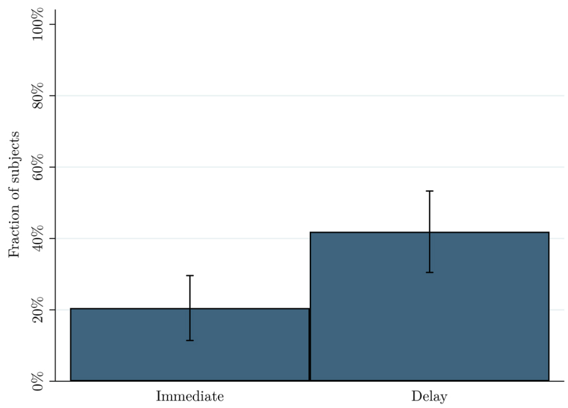
Table 3: Memory-based beliefs.

5.4 Figure 3: Suboptimal reinvestment; Table 4: Subjective memory and reinvestment decisions

Read: Delay increases suboptimal reinvestment, and individual memory bias predicts suboptimal reinvestment.

Detailed interpretation: The key outcome is not reinvestment per se, but reinvestment when the Bayesian benchmark says the bond is better. This shows that memory bias changes real choices.

Economic message: Investors repeat previously chosen investments too much when memory overweights gains.



(a) Figure 3: Suboptimal reinvestment.

Table 4
Subjective memory and reinvestment decisions

	(1) Inv.	(2) Inv. (subopt.)	(3) Inv. (subopt.)	(4) Inv. (subopt.)
Treatment (*Delay*)	2.213**	3.542***		
	(0.78)	(1.44)		
Memory bias (pos. out.)			1.599***	
			(0.22)	
Memory bias (neg. out.)				0.617***
				(0.10)
Constant	0.453	0.043***	0.077**	0.076**
	(0.27)	(0.05)	(0.08)	(0.08)
Session	Yes	Yes	Yes	Yes
N	152	152	152	152
Pseudo R^2	.05	.11	.13	.11

This table contains the odds ratios and standard errors (in parentheses) of logit regressions in which the dependent variables are a dummy variable, which is equal to one if the subject reinvested in the stock (*Inv.*), and a dummy variable, which is equal to one if the subject reinvested in the stock with a lower expected outcome than the bond (*Inv. (subopt.)*) after the observation phase (second choice) in experiment 1. *Treatment (Delay)* is a dummy variable representing our treatment with 1 = Delay condition and 0 = Immediate condition. *Memory bias (pos. out.)* represents subject's memory bias for observed positive outcomes. This variable is estimated at the individual level by subtracting the actual individually observed number of positive stock outcomes from subject's recalled number of the stock's positive outcomes. *Memory bias (neg. out.)* represents subject's memory bias for observed negative outcomes. This variable is estimated at the individual level by subtracting the actual individually observed number of negative stock outcomes from subject's recalled number of the stock's negative outcomes. We control for session fixed effects. * $p < .1$; ** $p < .05$; *** $p < .01$.

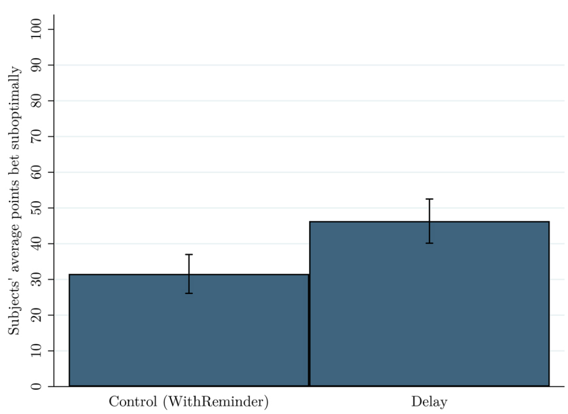
(b) Table 4: Subjective memory and reinvestment decisions.

5.5 Figure 4: Overconfidence: Suboptimal betting; Table 5: Subjective memory and confidence

Read: Delay increases points bet on the optimality of one's choice, including when the choice was suboptimal. Memory bias predicts suboptimal betting.

Detailed interpretation: The confidence result extends the mechanism from asset optimism to self-assessment. Biased memory protects the belief that the investor made a good decision.

Economic message: Positive memory bias can generate investor overconfidence.



(a) Figure 4: Overconfidence: Suboptimal betting.

Table 5
Subjective memory and confidence

	(1) Points bet	(2) Points bet (subopt.)	(3) Points bet (subopt.)
Treatment (*Delay*)	9.381***	14.788***	
	(2.85)	(4.18)	
Memory bias (pos. out.)			2.790**
			(1.25)
Constant	47.714***	31.537***	41.990***
	(1.96)	(2.75)	(3.63)
N	487	191	83
R^2	.02	.06	.06

This table contains the coefficients and standard errors (in parentheses) of OLS regressions in which the dependent variable indicates subjects' number of points bet on the optimality of their investment choice (*Points bet*) and a variable indicating subjects' number of points bet on the optimality of their investment choice despite having observed less than 6 positive outcomes (*Points bet (subopt.)*) in experiment 3. *Treatment (Delay)* is a dummy variable representing our treatment with 1 = Delay condition and 0 = Reminder condition. *Memory bias (pos. out.)* represents subject's memory bias for observed positive outcomes. This variable is estimated at the individual level by subtracting the actual individually observed number of positive stock outcomes from subject's recalled number of the stock's positive outcomes. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 5: Subjective memory and confidence.

5.6 Table 6: Subjective memory bias across experimental treatments; Table 7: The effect of high incentives on subjective memory bias for positive outcomes

Read: High accuracy incentives and removing active choice attenuate the memory bias. High incentives matter especially for subjects high in self-deceptive enhancement.

Detailed interpretation: These tables support motivated memory. The bias is weaker when accuracy is highly rewarded and when the subject did not actively choose the asset.

Economic message: Memory bias is not just passive forgetting; it is partly tied to self-image and choice responsibility.

Table 6
Subjective memory bias across experimental treatments

Treatment	Mean memory bias (pos. out.) in *Delay* condition	Treatment effect	Mechanism
Baseline (N = 185)	0.36 (p = .023)	0.48 (p = .010)	–
HighStakes (N = 154)	0.11 (p = .591)	0.32 (p = .151)	Memory suppression
NoChoice (N = 159)	0.18 (p = .398)	0.49 (p = .051)	Active choice

This table displays subjects' mean memory bias for positive outcomes in the *Delay* group, separately for subjects in the *Baseline*, *HighStakes*, and *NoChoice* condition as well as the treatment effect (the difference in memory bias for positive outcomes in the *Delay* group versus *Immediate* group) in experiment 2. Memory bias is estimated at the individual level by subtracting the actual individually observed number of positive outcomes from subject's recalled number of the stock's positive outcomes. The table reports mean values and t-test results against the null hypothesis that the memory bias is zero as well as the t-test results of the difference in means between *Delay* and *Immediate*.

(a) Table 6: Subjective memory bias across experimental treatments.

Table 7
The effect of high incentives on subjective memory bias for positive outcomes

	(1) Memory bias - High SDE	(2) Memory bias - Low SDE
Treatment (*HighStakes*)	–0.469**	0.115
	(0.19)	(0.22)
Constant	0.182	0.058
	(0.12)	(0.16)
N	172	167
R^2	.04	.00

This table contains the coefficients and standard errors (in parentheses) of OLS regressions in which the dependent variable is subject's memory bias for observed positive outcomes *Memory bias* in experiment 2. The two regression models restrict the sample to subjects who score above-average on the self-deceptive enhancement (SDE) scale (1) and to subjects who score below-average on the self-deceptive enhancement (SDE) scale (2). *Treatment (HighStakes)* is a dummy variable that equals one if the subject took part in the *HighStake* condition and zero if the subject took part in the *Baseline* condition. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 7: The effect of high incentives on subjective memory bias for positive outcomes.

6 Short summary

- Investors remember gains too much and losses too little after a delay.
- Biased memories produce optimistic beliefs relative to Bayesian posteriors.
- Memory bias leads to suboptimal reinvestment and overconfidence.
- High recall incentives and no-choice treatments point to motivated memory as a mechanism.

7 Conclusion

7.1 Core conclusion

Investor memory is an active part of learning. Investors do not simply update on past outcomes; they first remember those outcomes, and the remembered data are systematically tilted toward gains.

7.2 Economic intuition

Remembering gains supports a positive view of the investor's original choice, while remembering losses threatens that self-image. Over time, recall becomes selectively optimistic.

7.3 Contribution to behavioral finance

The paper provides a cognitive microfoundation for why investors learn more from successes than failures, repeat winning actions, and become overconfident after personal experience.

7.4 Limitations and next steps

The evidence is experimental and uses simplified investments. Future work could test the role of brokerage statements, financial advisers, complex portfolios, and professional investors.

7.5 Takeaway

Investor memory is not neutral storage. It is a biased filter that shapes beliefs, portfolio choices, and confidence.